

Retail Category Intelligence

Retail Product Performance — Exploratory Data Analysis

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Batch : 14
Analysis period • 01 January 2026 – 30 June 2026

107,836	₹459.80M	25	12
Transactions	Recorded revenue	Products	Stores

Report purpose

A structured view of category movement, product contribution, discount patterns and sales timing, translated into practical points for a category-manager review.

Design note

This edition uses a clean editorial layout with clear section breaks, compact tables and presentation-ready visuals so each analytical point can be read independently.

The six-month review covers 107,836 transactions and ₹459,804,176.78 in recorded revenue. Electronics contributes the largest share of revenue at 49.27%, but its January-to-June revenue fell by 35.0%. Apparel moved in the opposite direction, increasing by 34.1%. Grocery shows the lowest monthly coefficient of variation, making it the most consistent category in the period.

At product level, Smart TV is the lowest-revenue product within Electronics at ₹41,882,166.21. Saree and T-Shirt are the two highest-revenue products in Apparel. Across discount brackets, average revenue per transaction falls 22.5% between 0% and 20% discount, while average units sold stay almost unchanged. This is an observed association, not a causal result, and margin impact cannot be calculated from the available fields.

Key takeaways at a glance

Observation	What the data shows
Category movement	Electronics: -35.02% from January to June; Apparel: +34.10%.
Consistency	Grocery has the lowest monthly CV at 3.87%.
Product signal	Smart TV is the lowest-revenue Electronics product; Saree and T-Shirt lead Apparel.
Discount pattern	Average revenue/transaction falls from ₹4,685.65 at 0% to ₹3,631.73 at 20%.
Timing	Saturday has the highest chain-level average daily revenue at ₹3,523,605.44.

The source brief describes a regional retail chain with 12 stores across Andhra Pradesh and five categories. The Category Manager is preparing for a quarterly supplier review and needs evidence on category revenue trends, product-level strengths and weaknesses, discount effectiveness, day-of-week/month patterns and a clear category-level action.

CBIM Problem Canvas

Element	Working definition for this analysis
Situation (S)	A 12-store retail chain is preparing for a quarterly supplier review across five categories.
Complication (C)	The manager needs to separate growing, stable and declining categories and identify product-level drivers and weak spots.
Question (Q)	Which categories are changing, which products drive/drag revenue, how discounts relate to transaction revenue, and when sales are expected to change?
SSOT	The supplied MUC01 retail sales transaction dataset.
Success definition	A category priority, a specific challenge supported by numbers, a discount finding, a timing signal and clear limitations.
Primary stakeholder	The Category Manager preparing for the supplier review.

How the analysis flows

Extract	Transform	Validate	Analyse	Interpret
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The notebook loads the raw retail transactions into Pandas, standardises text and numeric fields, parses dates, validates business rules and creates analytical features. The cleaned dataframe is used for the EDA, with a cleaned CSV retained for reproducibility.

Validation check	Result
Rows loaded	107,836
Rows after cleaning	107,836
Duplicate transaction IDs	0
Rows with at least one null	0
Critical validation issues	0
Revenue mismatches > ₹0.01	0
Unique products	25
Unique categories	5
Unique stores	12
Unique cities	6

Revenue was reconciled against $\text{units_sold} \times \text{unit_price} \times (1 - \text{discount_pct}/100)$, using a ₹0.01 tolerance for stored rounding. No material mismatches were found, so the cleaning stage is primarily validation and feature transformation.

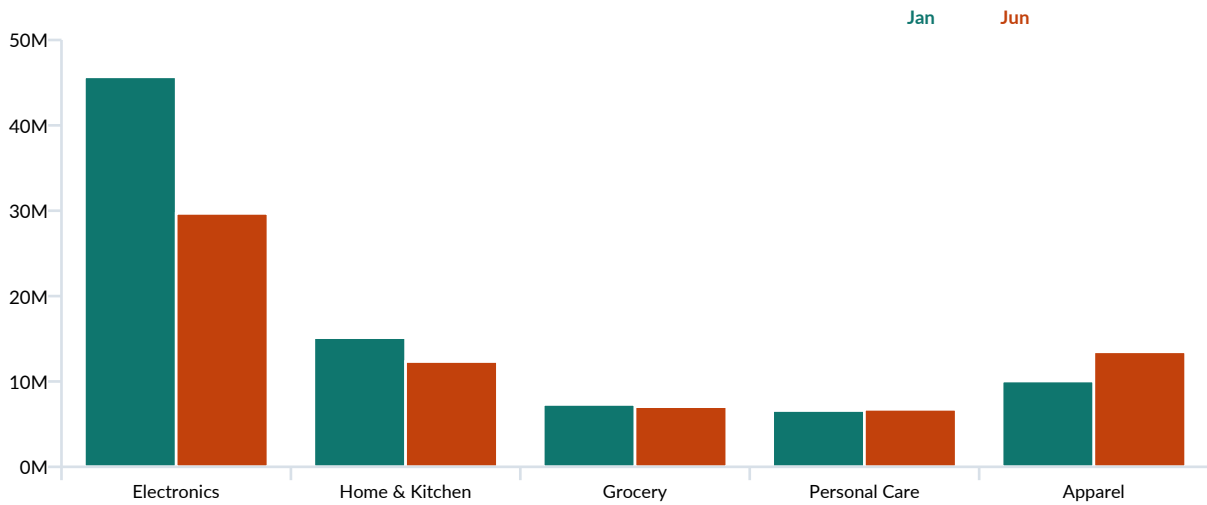
Dataset profile

Metric	Value
Transactions	107,836
Total revenue	₹459,804,176.78
Total units sold	278,593
Date range	01 Jan 2026 – 30 Jun 2026
Products	25
Categories	5
Stores	12
Cities	6

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Category Revenue Trend

January vs June revenue by category



Category	Jan revenue	Jun revenue	Change	Change %	Monthly CV
Electronics	₹45,686,596.63	₹29,686,824.81	₹-15,999,771.82	-35.02%	14.60%
Home & Kitchen	₹15,130,871.54	₹12,333,283.07	₹-2,797,588.47	-18.49%	7.26%
Grocery	₹7,307,011.34	₹7,047,719.23	₹-259,292.11	-3.55%	3.87%
Personal Care	₹6,593,734.81	₹6,737,354.12	₹143,619.31	2.18%	7.52%
Apparel	₹10,049,916.78	₹13,476,952.88	₹3,427,036.10	34.10%	13.88%

Reading the result: Electronics shows the largest decline, while Apparel records the strongest positive change. Grocery is comparatively stable because its monthly variation is the lowest.

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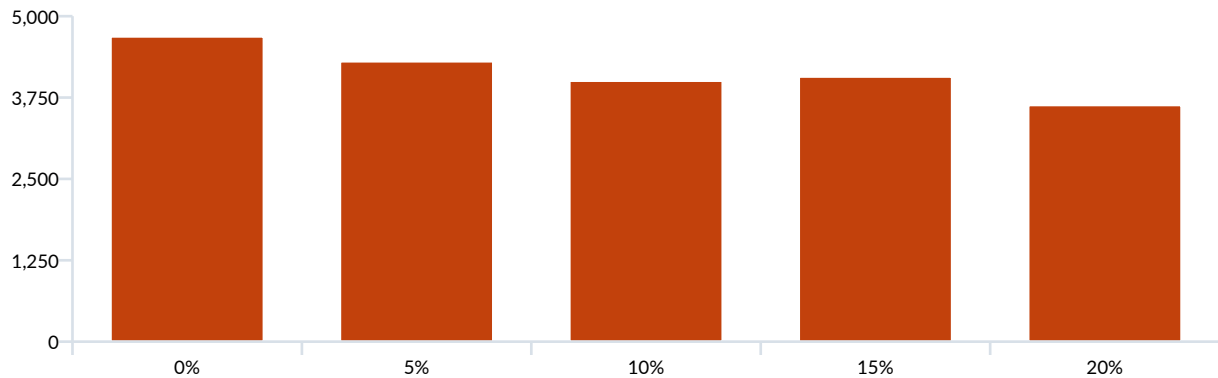
Product Contribution & Discount Pattern

Category	Top product	Revenue	Second product	Revenue	Lowest product	Revenue
Apparel	Saree	₹14,161,864.63	T-Shirt	₹14,099,251.65	Kurta	₹13,606,747.03
Electronics	Headphones	₹47,696,053.99	Laptop	₹46,551,078.39	Smart TV	₹41,882,166.21
Grocery	Rice (5kg)	₹8,562,171.86	Dal (1kg)	₹8,488,303.64	Sugar (1kg)	₹8,377,786.05
Home & Kitchen	Mixer Grinder	₹16,903,254.58	Storage Box	₹16,676,535.31	Water Bottle	₹15,825,842.17
Personal Care	Shampoo	₹8,168,328.26	Body Lotion	₹8,106,578.00	Sunscreen	₹7,593,863.45

The product view adds context to the category trend: Electronics has Smart TV as its lowest-revenue product, while Apparel's leading products are Saree and T-Shirt.

Discount effectiveness

Average revenue per transaction by discount bracket



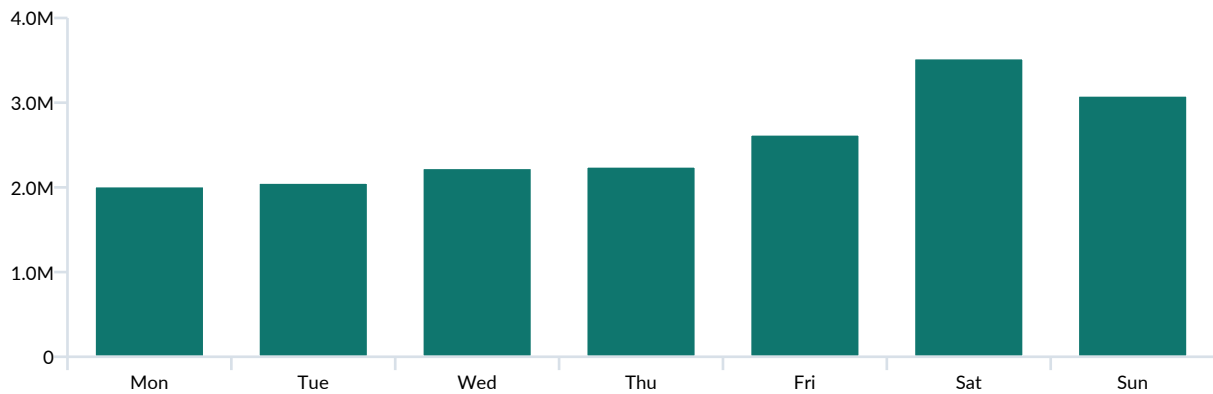
Discount	Transactions	Avg revenue / transaction	Avg units
0%	36,033	₹4,685.65	2.59
5%	23,922	₹4,303.73	2.59
10%	23,810	₹4,005.55	2.57
15%	11,935	₹4,069.16	2.60
20%	12,136	₹3,631.73	2.58
25%+	0	—	—

From 0% to 20% discount, average revenue per transaction is 22.5% lower, while average units sold move only from 2.59 to 2.58. The relationship is observational; the dataset does not include product cost or gross margin, and the 25%+ bracket has no observations.

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Sales Timing: Weekday & Monthly Pattern

Average daily revenue by weekday — chain overall



Day	Average daily revenue
Monday	₹2,012,140.96
Tuesday	₹2,054,129.70
Wednesday	₹2,228,117.24
Thursday	₹2,245,493.02
Friday	₹2,623,611.89
Saturday	₹3,523,605.44
Sunday	₹3,083,374.58

Timing signal: Saturday has the highest average daily revenue, while Tuesday is the lowest at chain level.

Monthly chain pattern

The report identifies January as the strongest month at ₹84,768,131.10 and June as the weakest at ₹69,282,134.11.

Electronics timing

For Electronics, Saturday is the strongest weekday at ₹1,786,658.45 average daily revenue and Tuesday is the weakest at ₹970,975.52. Electronics also peaks in January at ₹45,686,596.63 and reaches its lowest monthly total in June at ₹29,686,824.81.

- Apparel shows a strong positive movement from January to June, supported by Saree and T-Shirt at product level.
- Electronics needs closer supplier-review attention because revenue declined 35.0%, with Smart TV the lowest-revenue product in the category.
- The discount results do not show a matching increase in average units sold as discounts rise; deeper discounting therefore needs evidence of incremental demand.
- Saturday is the strongest average-revenue day for both the overall chain and Electronics, making it a useful timing signal for promotion experiments.
- Before making a profitability decision, additional cost/margin, inventory/stock-out and promotion-exposure information would be needed.

Action plan

Area	Suggested action
Investment test	Run controlled incremental shelf/promotion testing around Apparel.
Supplier review	Review Electronics' January-to-June decline and Smart TV contribution.
Discount review	Ask for evidence of incremental demand before moving to deeper discounts.
Timing	Use Saturdays as the main timing signal; test weaker days separately.
Data request	Add cost/margin, inventory/stock-out and promotion-exposure fields.

Important limitations

- The dataset does not contain product cost or gross margin, so profitability cannot be calculated.
- Discount brackets are observational; product mix, store mix and promotion targeting may influence the averages.
- The 25%+ discount bracket has no observations.
- The analysis covers only January–June 2026, so longer-term seasonality cannot be assessed.
- Inventory, stock-out, campaign-exposure and customer-level variables are not available.

Deliverables

The analysis package consists of the end-to-end notebook, a submission-ready insight summary, this detailed report and the cleaned CSV supporting the ETL stage.

Final note

The report separates the observed numbers from interpretation and keeps the limitations visible so that the findings can be reviewed in context.